

XIAMEN UNIVERSITY MALAYSIA

厦門大學馬來西亞分校

COM105 Engineering Physics (I)

Lecture 13: The Second Law of Thermodynamics

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In this chapter, you'll learn…

- the difference between reversible (可逆) and irreversible (不可逆) processes.
- the physics of internal-combustion engines.
- how refrigerators and heat engines are related, and how to analyze the performance of a refrigerator.
- how the second law of thermodynamics sets limits on the efficiency of engines and the performance of refrigerators.

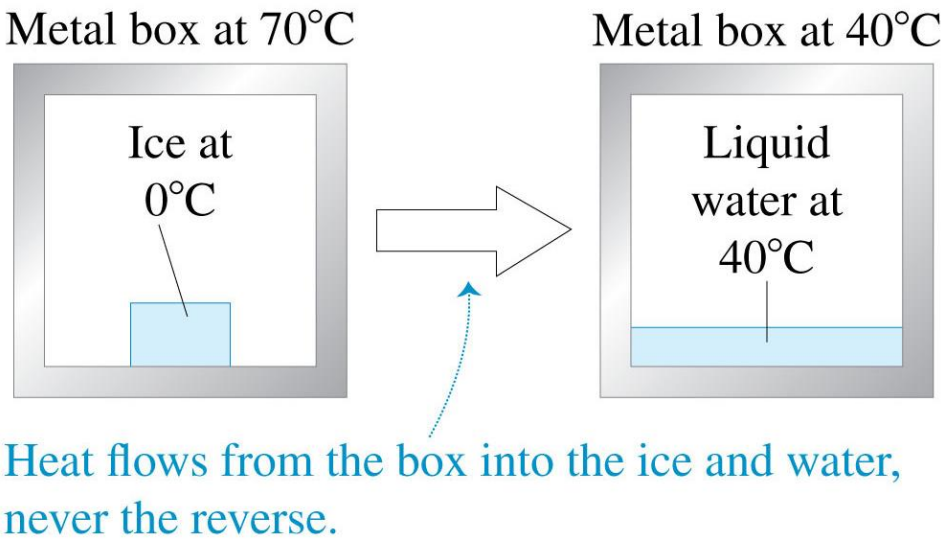
Directions of Thermodynamic Processes

- Why does heat flow from the hot lava into the cooler water?
- Could it flow the other way?
- It is easy to convert mechanical energy completely into heat, but not the reverse. Why not?
- We need to use the second law of thermodynamics and the concept of entropy to answer the above questions.



Directions of Thermodynamic Processes

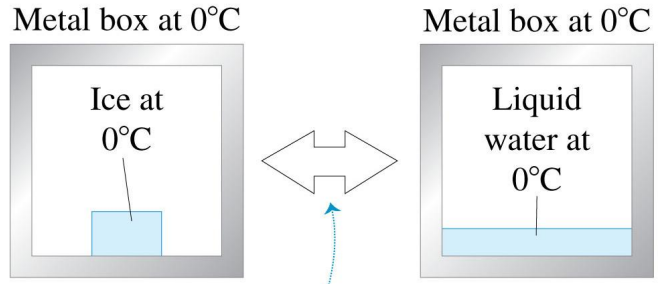
- The direction of a reversible process can be reversed by an infinitesimal change in its conditions.
- The system is always in or very close to thermal equilibrium.
- For example, a block of ice melts irreversibly when we place it in a hot metal box.



Directions of Thermodynamic Processes

- A system that undergoes such an idealized reversible process is always very close to being in thermodynamic equilibrium within itself and with its surroundings.
- For example, a block of ice at 0°C can be melted reversibly if we put it in a 0°C metal box.

  - Idealized reversible processes are equilibrium processes.
  - Irreversible processes are nonequilibrium processes.
  - The second law of thermodynamics determines the preferred direction for such processes.
  - To introduce the second law of thermodynamics by considering two broad classes of devices: heat engines, and refrigerator.



By infinitesimally raising or lowering the temperature of the box, we can make heat flow into the ice to melt it or make heat flow out of the water to refreeze it.

Heat Engines

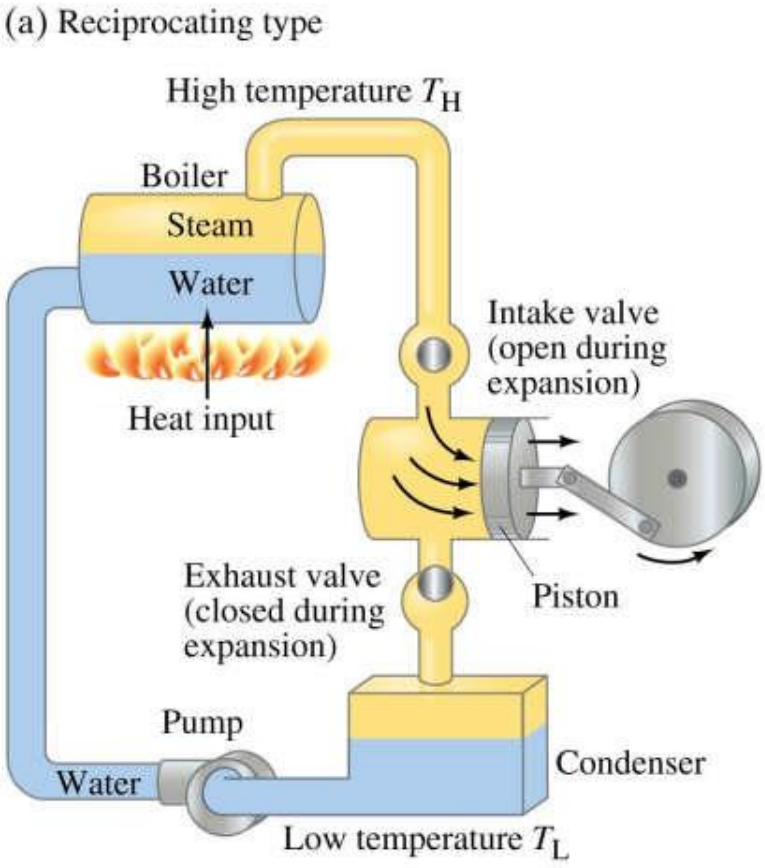
- A heat engine is any device that partly transforms heat into work or mechanical energy.
- All motorized vehicles other than purely electric vehicles use heat engines for propulsion.推进
- (Hybrid vehicles use their internal-combustion engine to help charge the batteries for the electric motor.)

内燃



Heat Engines 热机

- The heat engine carries the working substance (usually gasoline or diesel) through a cyclic process, during which:
  - the working substance absorbs heat from a hot reservoir
  - part of heat absorbed is converted into mechanical energy
  - remaining heat is expelled to a low temperature reservoir



(Steam Engine)

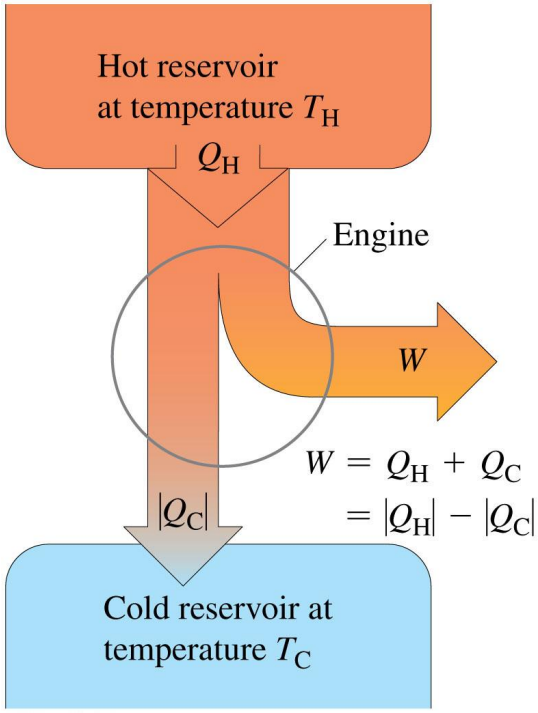
Heat Engines

- Simple heat engines operate on a cyclic process during which they absorb heat Q_H from a hot reservoir, perform some mechanical work, and discard some heat Q_C to a cold reservoir.
- The heat engine repeats the same cycle over and over, Q_H and Q_C represent the quantities of heat absorbed and rejected by the engine during one cycle; Q_H is positive, and Q_C is negative.
- The net heat Q absorbed per cycle is

$$Q = Q_H + Q_C = |Q_H| - |Q_C| \quad (20.1)$$

- The useful output of the engine is the net work W done by the working substance

$$W = Q = Q_H + Q_C = |Q_H| - |Q_C| \quad (20.2)$$



Schematic energy-flow diagram for a heat engine

Q_C = 耗散

The Efficiency of a Heat Engine

- Ideally, to convert all the heat into work:

$$Q_H = W \text{ and } Q_C = 0$$

- But this is impossible; there is always some heat wasted, and is never zero.
- The thermal efficiency *e* of a heat engine is the fraction of Q_H that is converted to work.

Thermal efficiency of an engine

$$e = \frac{W}{Q_H} = 1 + \frac{Q_C}{Q_H} = 1 - \left|\frac{Q_C}{Q_H}\right| \quad (20.4)$$

Work done by engine; Heat rejected by engine; Heat absorbed by engine

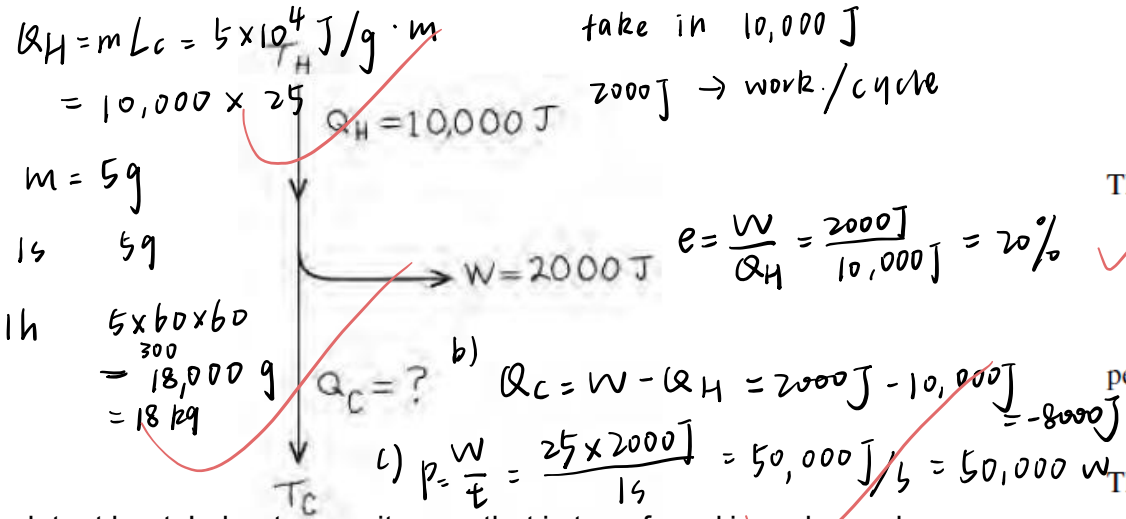
$$W = Q = Q_H + Q_C$$

- *e* is what you get divided by what you pay for.
- This is always less than unity, an all-too-familiar experience!

The Efficiency of a Heat Engine

EXAMPLE 20.1 ANALYZING A HEAT ENGINE

A gasoline truck engine takes in 10,000 J of heat and delivers 2000 J of mechanical work per cycle. The heat is obtained by burning gasoline with heat of combustion $L_c = 5.0 \times 10^4$ J/g. (a) What is the thermal efficiency of this engine? (b) How much heat is discarded in each cycle? (c) If the engine goes through 25 cycles per second, what is its power output in watts? In horsepower? (d) How much gasoline is burned in each cycle? (e) How much gasoline is burned per second? Per hour?



latent heat, L, heat per unit mass that is transferred in a phase change

EXECUTE: (a) From Eq. (20.4), the thermal efficiency is

$$e = \frac{W}{Q_H} = \frac{2000\text{ J}}{10{,}000\text{ J}} = 0.20 = 20\%$$

(b) From Eq. (20.2), $W = Q_H + Q_C$, so

$$Q_C = W - Q_H = 2000\text{ J} - 10{,}000\text{ J} = -8000\text{ J}$$

That is, 8000 J of heat leaves the engine during each cycle.

(c) The power P equals the work per cycle multiplied by the number of cycles per second:

$$P = (2000\text{ J/cycle})(25\text{ cycles/s}) = 50{,}000\text{ W} = 50\text{ kW}$$

$$= (50{,}000\text{ W})\frac{1\text{ hp}}{746\text{ W}} = 67\text{ hp}$$

(d) Let m be the mass of gasoline burned during each cycle. Then $Q_H = mL_c$ and

$$m = \frac{Q_H}{L_c} = \frac{10{,}000\text{ J}}{5.0 \times 10^4\text{ J/g}} = 0.20\text{ g}$$

(e) The mass of gasoline burned per second equals the mass per cycle multiplied by the number of cycles per second:

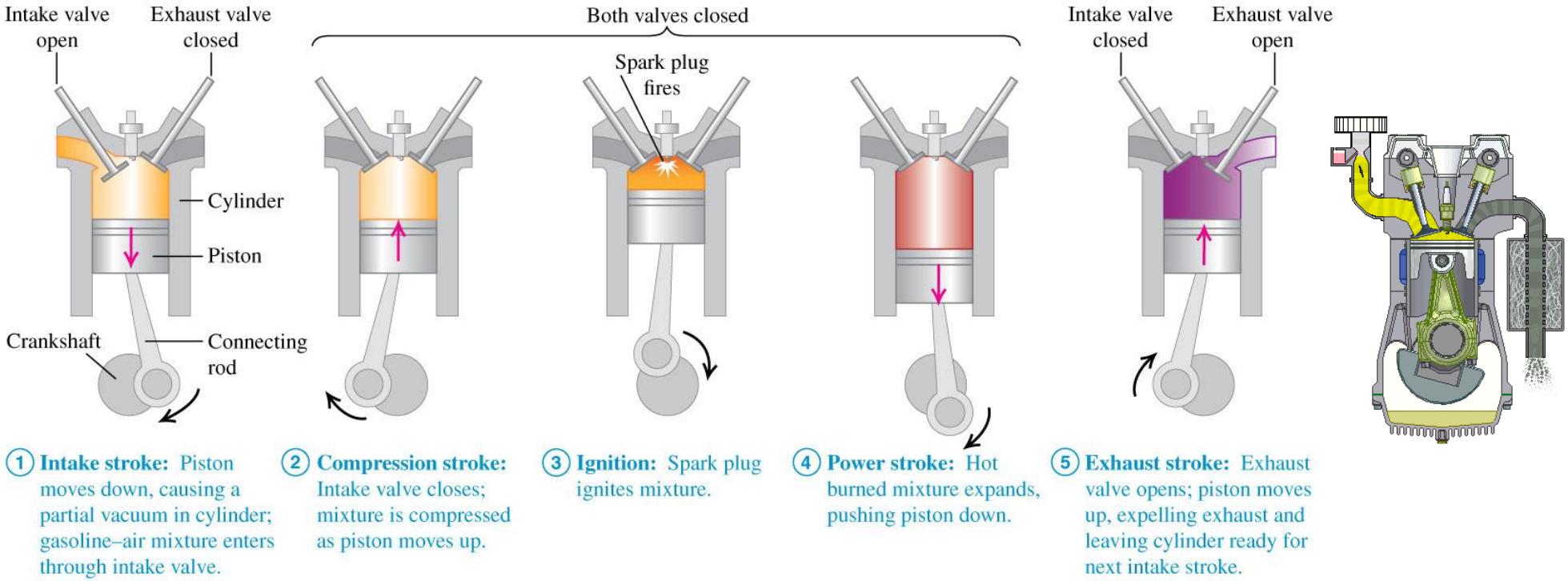
$$(0.20\text{ g/cycle})(25\text{ cycles/s}) = 5.0\text{ g/s}$$

The mass burned per hour is

$$(5.0\text{ g/s})\frac{3600\text{ s}}{1\text{ h}} = 18{,}000\text{ g/h} = 18\text{ kg/h}$$

Internal-Combustion Engines 内燃机

- Generates motive power by the burning of petrol, oil, or other fuel with air inside the engine, the hot gases produced being used to drive a piston or do other work as they expand.

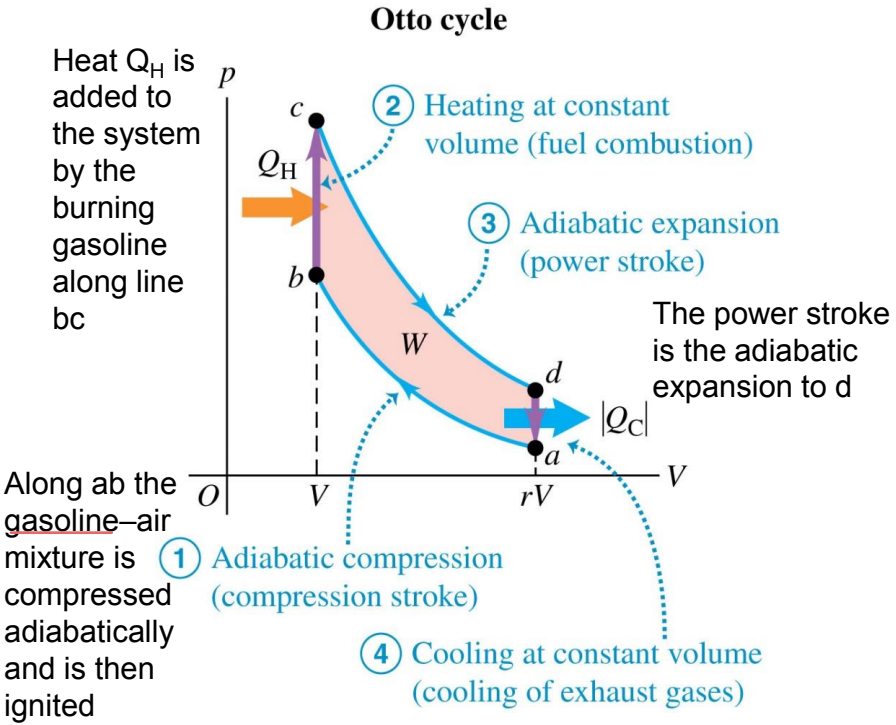


Compression ratio r: minimum volume V, maximum volume rV.

— present-day automobile engine: r is 8~10

Otto Cycle

- An Otto cycle is an idealized thermodynamic cycle that describes the functioning of a typical internal-combustion engine with spark ignition piston. 电火花活塞



- Processes bc and da are constant-volume, so the heats are related simply to the temperatures:

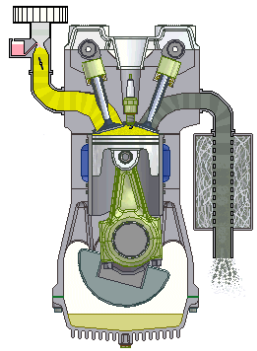
$$Q_H = nC_V(T_c - T_b) > 0$$

$$Q_C = nC_V(T_a - T_d) < 0$$

- Efficiency:

$$e = \frac{Q_H + Q_C}{Q_H} = \frac{T_c - T_b + T_a - T_d}{T_c - T_b}$$

(20.5)



对两个过程，ab. cd.

Otto Cycle

- To further simplify by using the temperature–volume relationship for adiabatic processes for an ideal gas. For the two adiabatic processes ab and cd (from Eq. 19.22).

$$T_a(rV)^{\gamma-1} = T_bV^{\gamma-1} \quad \text{and} \quad T_d(rV)^{\gamma-1} = T_cV^{\gamma-1}$$

- So, $e = \dfrac{T_dr^{\gamma-1} - T_ar^{\gamma-1} + T_a - T_d}{T_dr^{\gamma-1} - T_ar^{\gamma-1}} = \dfrac{(T_d - T_a)(r^{\gamma-1} - 1)}{(T_d - T_a)r^{\gamma-1}}$

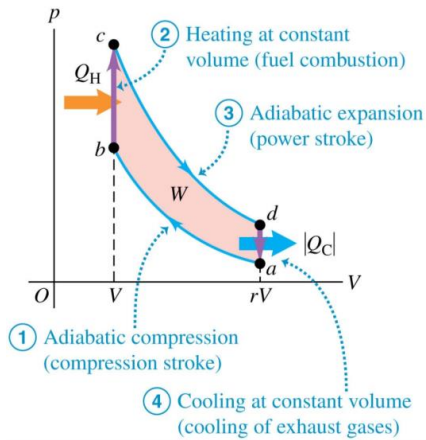
- Dividing out the common factor ($T_d - T_a$)

Thermal efficiency in Otto cycle

$$e = 1 - \frac{1}{r^{\gamma-1}}$$

Compression ratio; Ratio of heat capacities

(20.6)

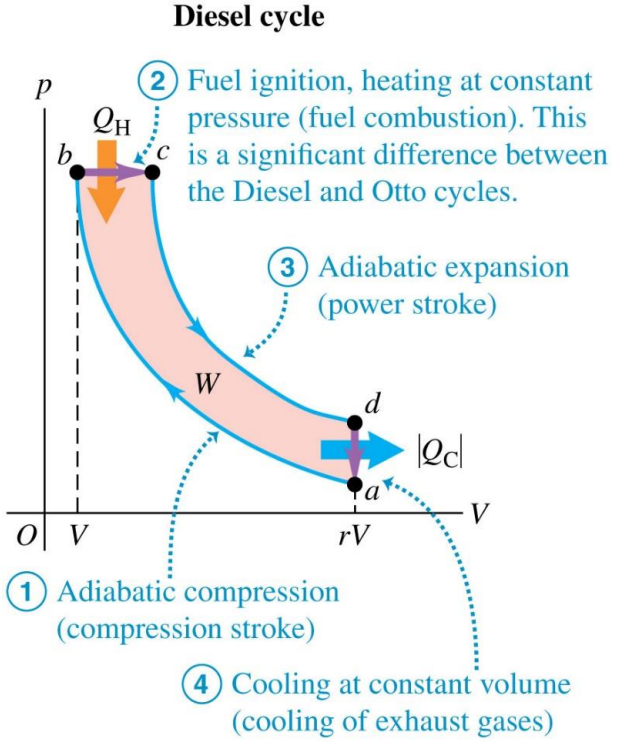


- With r = 8 and γ = 1.4 (the value for air) the theoretical efficiency is e = 0.56, or 56%. The efficiency can be increased by increasing r, but increasing r can damage the engine.
- The maximum practical compression ratio for high-octane, or “premium,” gasoline is about 10 to 13.

**The Otto cycle is a highly idealized model. It assumes that the mixture behaves as an ideal gas; it ignores friction, turbulence, loss of heat to cylinder walls, and many other effects that reduce the efficiency of an engine. Efficiencies of real gasoline engines are typically around 35%.*

The Diesel Cycle

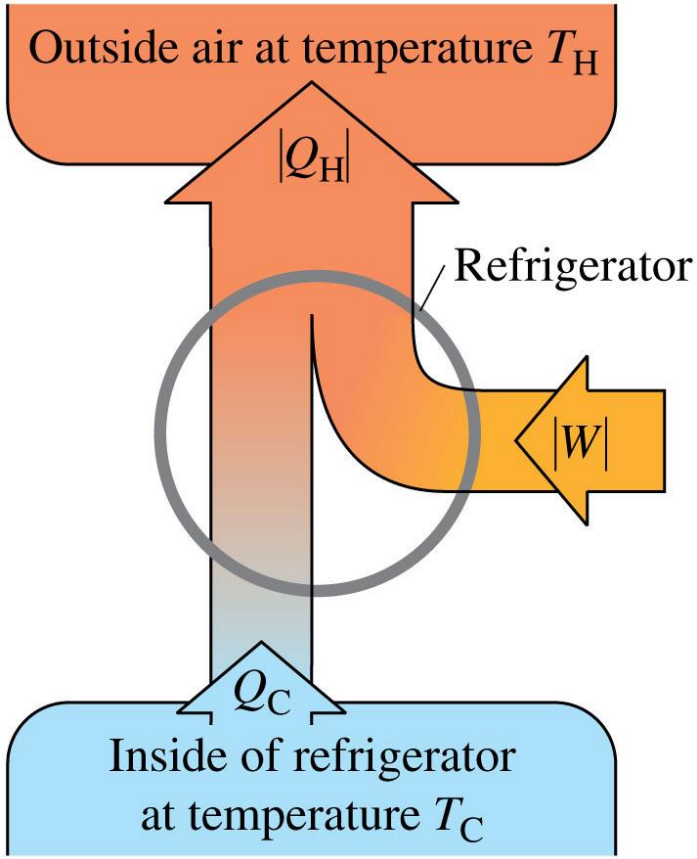
- Starting at point a, air is compressed adiabatically to point b, heated at constant pressure to point c, expanded adiabatically to point d, and cooled at constant volume to point a.
- Because there is no fuel in the cylinder during the compression stroke, pre-ignition cannot occur, and the compression ratio r can be much higher than for a gasoline engine.
- This improves efficiency and ensures reliable ignition.
- Values of r of 15 to 20 are typical.



- Theoretical efficiency of the idealized Diesel cycle is about 0.65 to 0.70.
- Much tighter tolerances than gasoline engines and the fuel-injection system requires careful maintenance.

Refrigerators

- A refrigerator takes heat from a cold place (inside the refrigerator) and gives it off to a warmer place (the room). An input of mechanical work is required to do this.
- A refrigerator is essentially a heat engine operating in reverse.
- Shown is an energy-flow diagram of a refrigerator.



$Q_C > 0$
$W < 0$ $Q_H < 0$

Refrigerators

- For a refrigerator, Q_C is positive but both W and Q_H are negative, so

$$|W| = -W \text{ and } |Q_H| = -Q_H$$

- From the first law for a cyclic process,

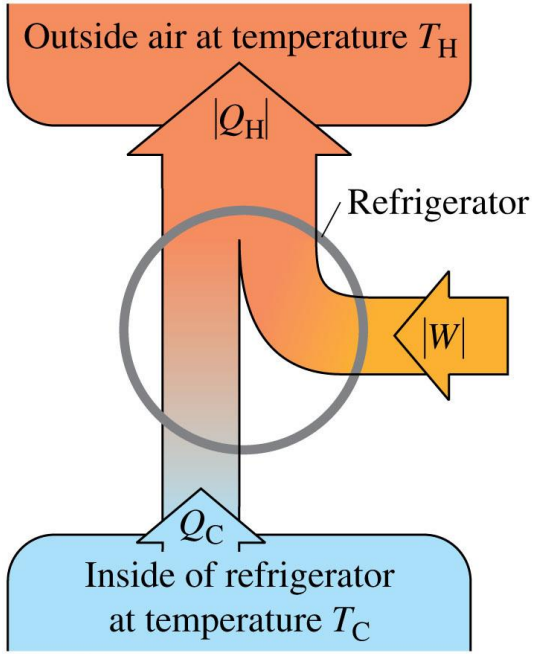
$$Q_H + Q_C - W = 0 \quad \text{or} \quad -Q_H = Q_C - W$$

- Therefore,

$$|Q_H| = Q_C + |W| \qquad (20.7)$$

- The absolute-value relationship:

$$|Q_H| = |Q_C| + |W| \qquad (20.8)$$



Refrigerators

- From an economic point of view, the best refrigeration cycle is one that removes the greatest amount of heat from the inside of the refrigerator for the least expenditure of mechanical work.
- The relevant ratio is therefore: $\frac{|Q_C|}{|W|}$
- We call this ratio the coefficient of performance, K:

Coefficient of performance of a refrigerator

$$K = \frac{|Q_C|}{|W|} = \frac{|Q_C|}{|Q_H| - |Q_C|} \qquad (20.9)$$

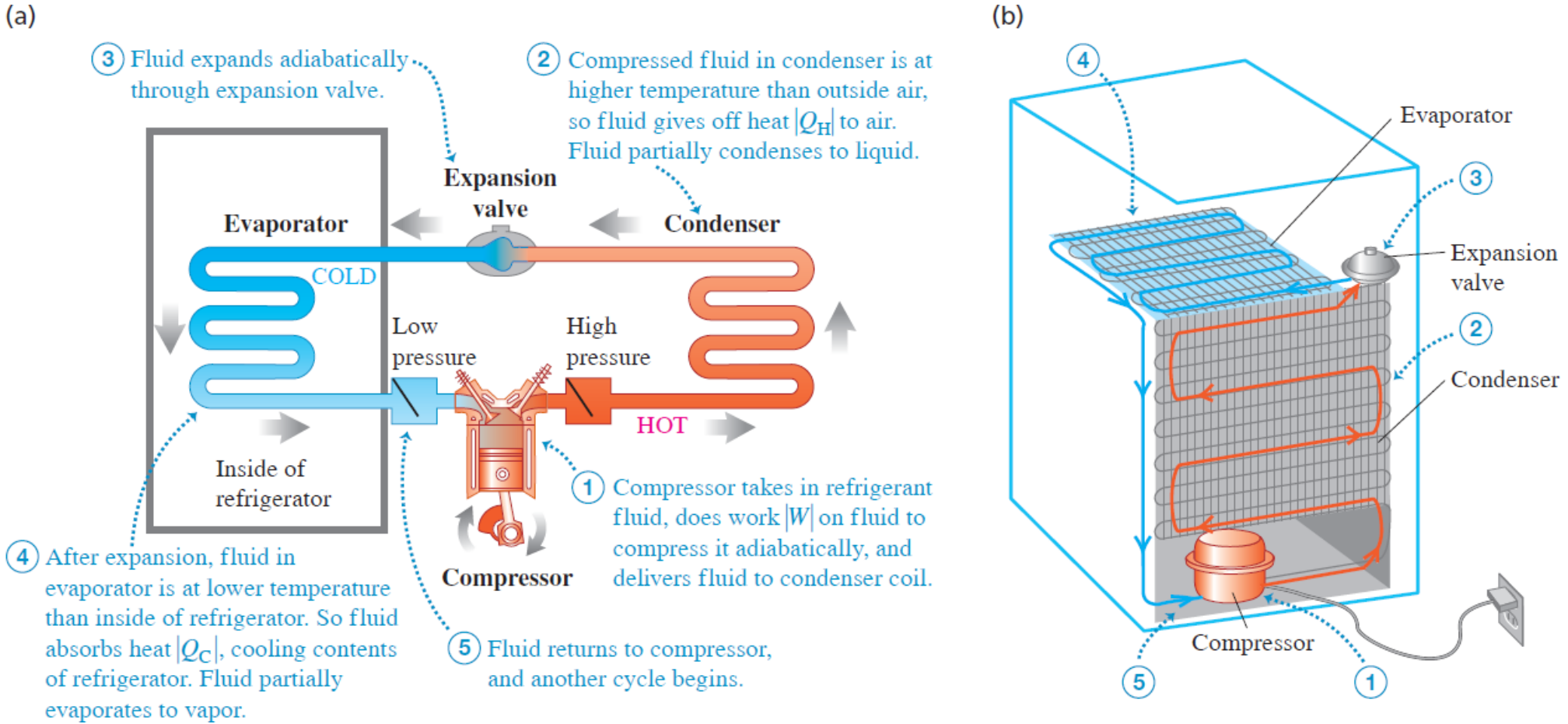
Heat removed from inside of refrigerator

Work input of refrigerator

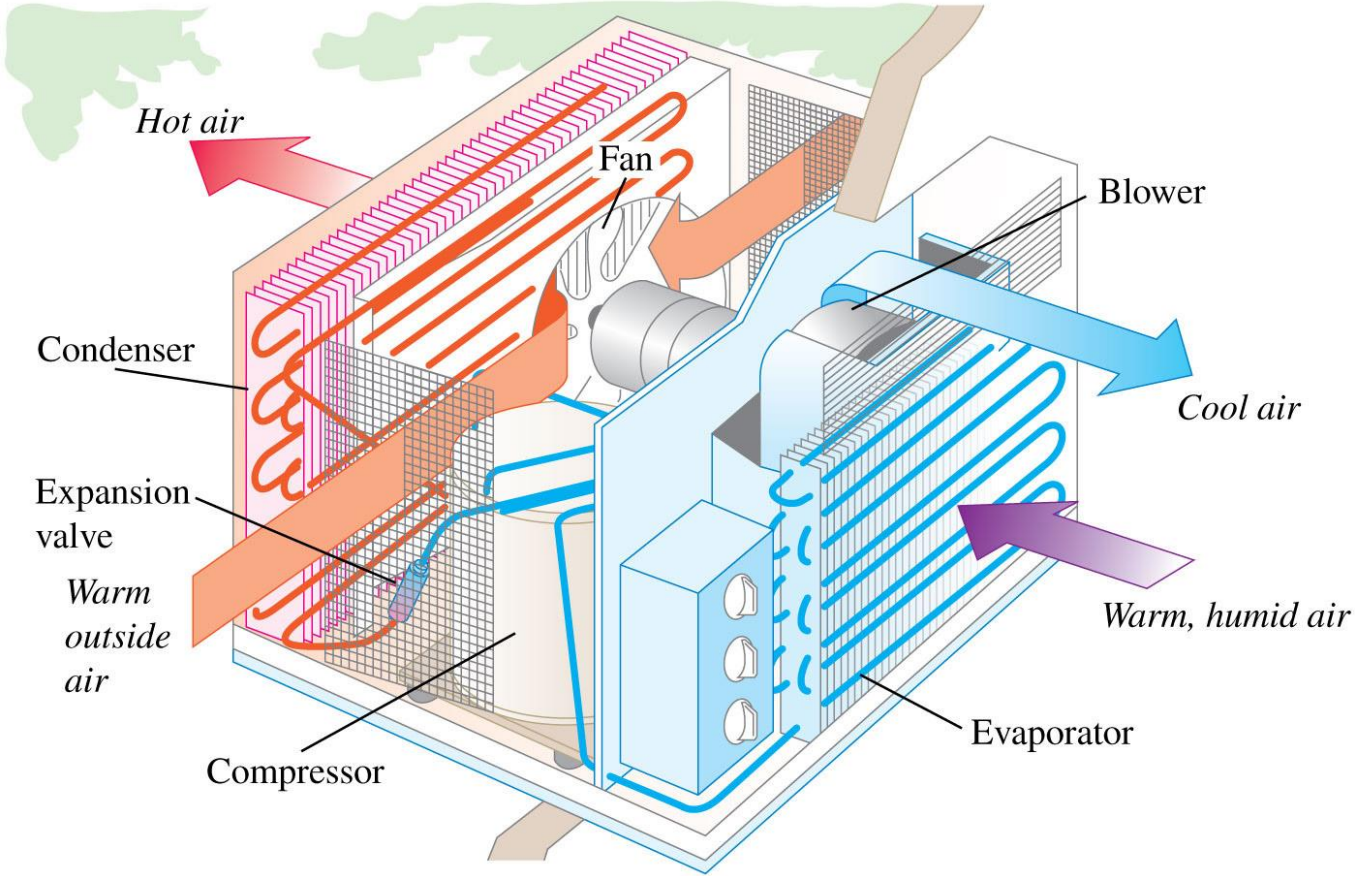
Heat rejected to outside air

- The larger the ratio, the better the refrigerator. The effectiveness of the refrigerator is given by its coefficient of performance K.

Refrigerators



Air Conditioner



The Second Law of Thermodynamics

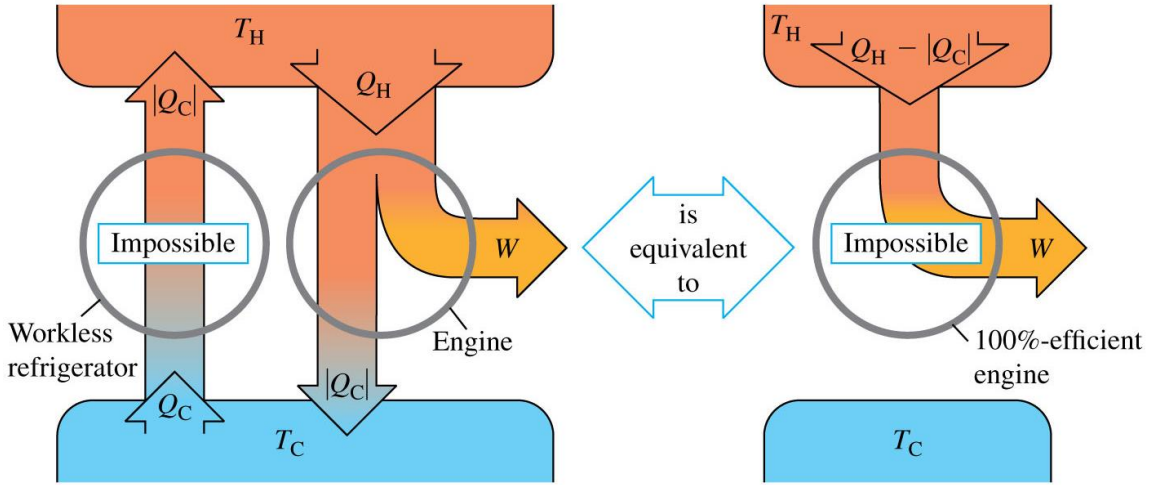
- The second law of thermodynamics can be stated in several ways:
  - ✓ It is impossible for any system to undergo a process in which it absorbs heat from a reservoir at a single temperature and converts the heat completely into mechanical work, with the system ending in the same state in which it began.
  - ✓ It is impossible for any process to have as its sole result the transfer of heat from a cooler to a hotter object.

We will call this the "engine" statement of the second law.

We'll call this the "refrigerator" statement of the second law.

The Second Law of Thermodynamics

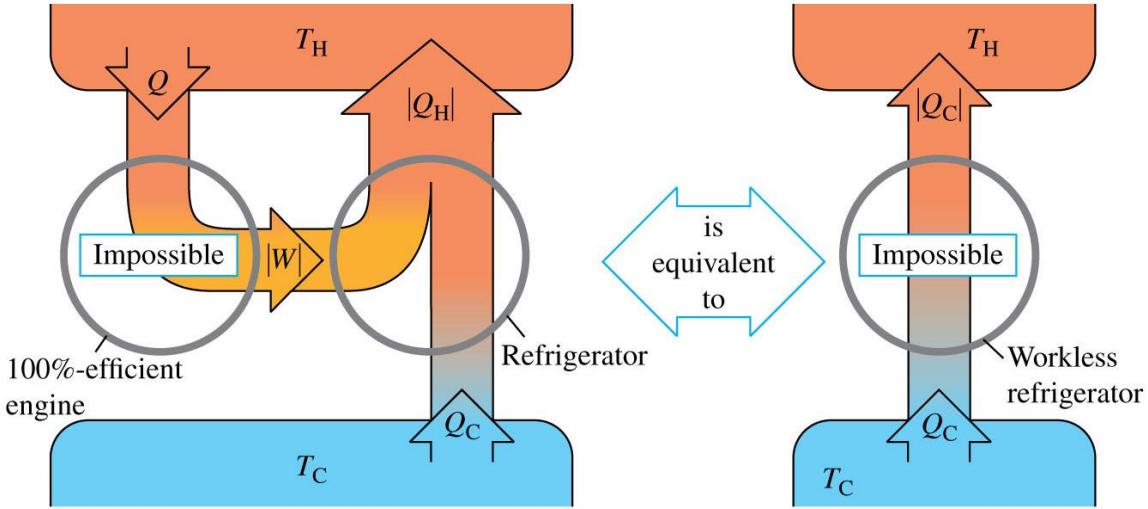
- The “engine” statement of the second law.



- If a workless refrigerator were possible, it could be used in conjunction with an ordinary heat engine to form a 100%-efficient engine, converting heat $Q_H - |Q_C|$ completely to work.

The Second Law of Thermodynamics

- The “refrigerator” statement of the second law.



- If a 100%-efficient engine were possible, it could be used in conjunction with an ordinary refrigerator to form a workless refrigerator, transferring heat Q_C from the cold to the hot reservoir with no input of work.

The Carnot Cycle

- According to the second law, no heat engine can have 100% efficiency. How great an efficiency can an engine have, given two heat reservoirs at temperatures T_H and T_C?
- This question was answered in 1824 by the French engineer Sadi Carnot, which was then known as Carnot Cycle:
  1. For maximum heat-engine efficiency, we must avoid all irreversible processes.
  2. The process involving heat transfer must be isothermal at either T_H or T_C
  3. The process in which the temperature T of the working substance changes must be adiabatic.

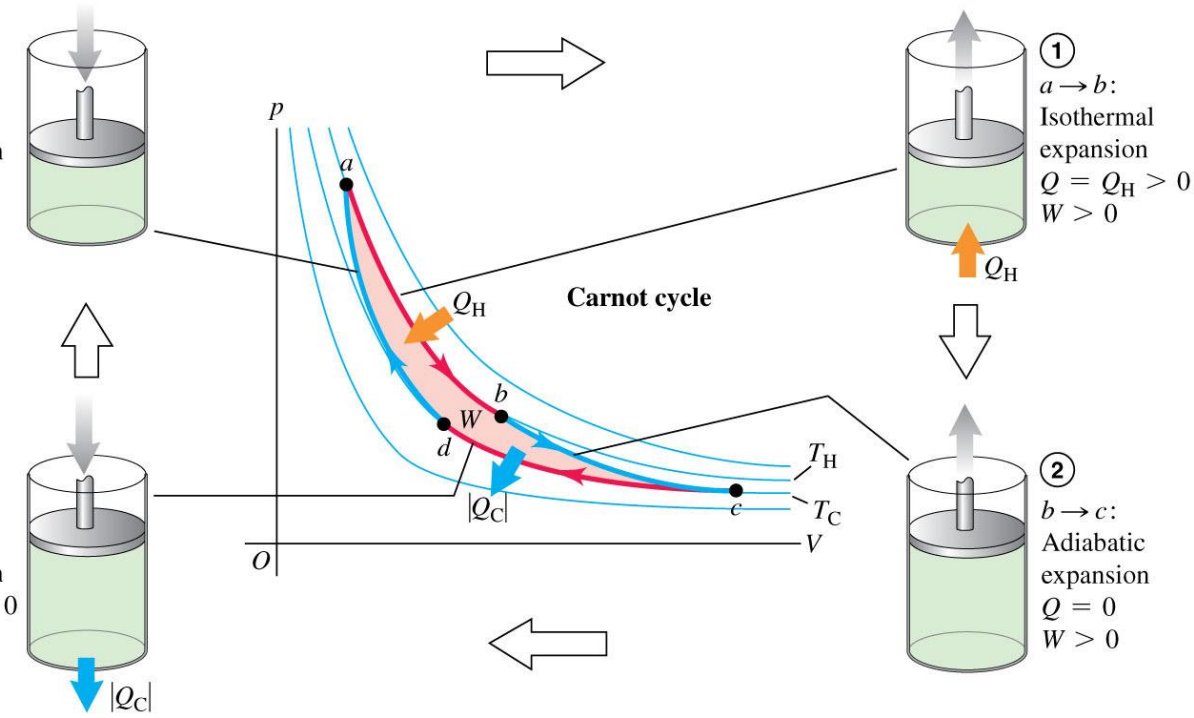
The Carnot Cycle

- A Carnot cycle has two adiabatic segments and two isothermal segments.

隔热 等温

It is compressed adiabatically back to its initial state at temperature T_H (da)

④ $d \rightarrow a$: Adiabatic compression $Q = 0$ $W < 0$



① $a \rightarrow b$: Isothermal expansion $Q = Q_H > 0$ $W > 0$

The gas expands isothermally at temperature T_H, absorbing heat Q_H (ab)

It is compressed isothermally at T_C, rejecting heat $|Q_C|$ (cd)

③ $c \rightarrow d$: Isothermal compression $Q = Q_C < 0$ $W < 0$

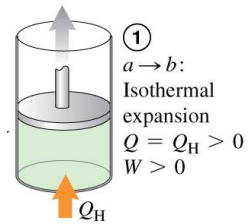
② $b \rightarrow c$: Adiabatic expansion $Q = 0$ $W > 0$

It expands adiabatically until its temperature drops to T_C (bc)

The Carnot Cycle

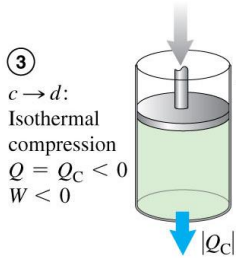
- Thermal Efficiency of Carnot Cycle
- For the isothermal expansion ab, $\Delta U_{ab} = 0$

$$Q_H = W_{ab} = nRT_H \ln \frac{V_b}{V_a} \qquad (20.10)$$



- For the isothermal compression cd,

$$Q_C = W_{cd} = nRT_C \ln \frac{V_d}{V_c} = -nRT_C \ln \frac{V_c}{V_d} \qquad (20.11)$$



- Because V_d is less than V_c , Q_C is negative (Q_C = - $|Q_C|$); heat flows out of the gas during the isothermal compression at temperature T_C.

$$\frac{Q_C}{Q_H} = -\left(\frac{T_C}{T_H}\right)\frac{\ln (V_c/V_d)}{\ln (V_b/V_a)} \qquad (20.12)$$

The Carnot Cycle

- Using the temperature–volume relationship for an adiabatic process: $T_H V_b^{\gamma-1} = T_C V_c^{\gamma-1}$ and $T_H V_a^{\gamma-1} = T_C V_d^{\gamma-1}$
- We then get: $\frac{V_b^{\gamma-1}}{V_a^{\gamma-1}} = \frac{V_c^{\gamma-1}}{V_d^{\gamma-1}}$ and $\frac{V_b}{V_a} = \frac{V_c}{V_d}$
- Hence, the heat transfer in a Carnot engine:

$$\frac{Q_C}{Q_H} = -\left(\frac{T_C}{T_H}\right)\frac{\ln(V_c/V_d)}{\ln(V_b/V_a)} \quad \Rightarrow \quad \frac{Q_C}{Q_H} = -\frac{T_C}{T_H} \text{ or } \frac{|Q_C|}{|Q_H|} = \frac{T_C}{T_H} \qquad (20.13)$$

- The efficiency of a Carnot engine is then:

Efficiency of a Carnot engine

$$e_{\text{Carnot}} = 1 - \frac{T_C}{T_H} = \frac{T_H - T_C}{T_H} \qquad (20.14)$$

(T_C: Temperature of cold reservoir; T_H: Temperature of hot reservoir)

(20.4)

Thermal efficiency of an engine

$$e = \frac{W}{Q_H} = 1 + \frac{Q_C}{Q_H} = 1 - \left|\frac{Q_C}{Q_H}\right|$$

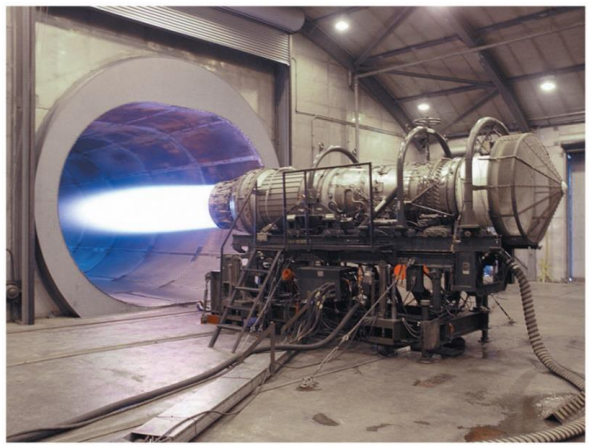
(W: Work done by engine; Q_C: Heat rejected by engine; Q_H: Heat absorbed by engine)

Carnot Cycle 要求 进温很高 出温很低

The Carnot Cycle

- This simple result says that the efficiency of a Carnot engine depends only on the temperatures of the two heat reservoirs. The efficiency is large when the temperature difference is large, and it is very small when the temperatures are nearly equal.
- To maximize the efficiency of a real engine, the designer must make the intake temperature T_H as high as possible and the exhaust temperature T_C as low as possible.
- For this reason, the temperatures inside a jet engine are made as high as possible.
- Exotic ceramic materials are used that can withstand temperatures in excess of 1000 degrees Celsius without melting or becoming soft.

陶瓷材料比较苛条件



The Carnot Cycle

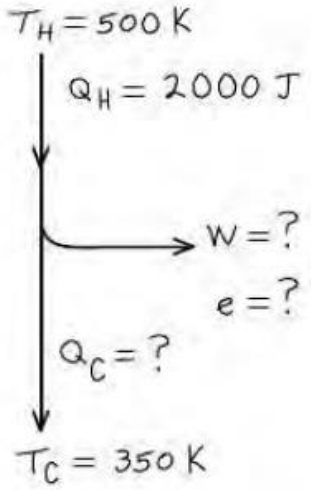
EXAMPLE 20.2 ANALYZING A CARNOT ENGINE I

A Carnot engine takes 2000 J of heat from a reservoir at 500 K, does some work, and discards some heat to a reservoir at 350 K. How much work does it do, how much heat is discarded, and what is its efficiency?

→2000J. at 500K.

→ work.

350K ↓



EXECUTE: From Eq. (20.13), $Q_C = -Q_H \frac{T_C}{T_H} = -1400\,J$

$$Q_C = -Q_H \frac{T_C}{T_H} = -(2000\ \text{J})\frac{350\ \text{K}}{500\ \text{K}} = -1400\ \text{J}$$

Then from Eq. (20.2), the work done is

$$W = Q_H + Q_C = 2000\ \text{J} + (-1400\ \text{J}) = 600\ \text{J}$$

From Eq. (20.14), the thermal efficiency is

$$e = 1 - \frac{T_C}{T_H} = 1 - \frac{350\ \text{K}}{500\ \text{K}} = 0.30 = 30\%$$

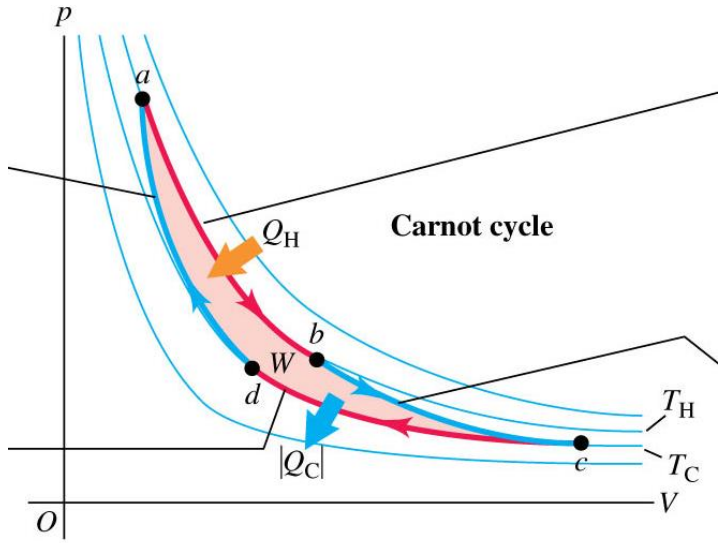
EVALUATE: The negative sign of Q_C is correct: It shows that 1400 J of heat flows *out* of the engine and into the cold reservoir. We can check our result for e by using the basic definition of thermal efficiency, Eq. (20.3):

$$e = \frac{W}{Q_H} = \frac{600\ \text{J}}{2000\ \text{J}} = 0.30 = 30\%$$

The Carnot Cycle

EXAMPLE 20.3 ANALYZING A CARNOT ENGINE II

Suppose 0.200 mol of an ideal diatomic gas ($\gamma = 1.40$) undergoes a Carnot cycle between 227°C and 27°C, starting at $p_a = 10.0 \times 10^5$ Pa at point a in the pV-diagram of Fig. 20.13. The volume doubles during the isothermal expansion step $a \rightarrow b$. (a) Find the pressure and volume at points a, b, c, and d. (b) Find Q, W, and ΔU for each step and for the entire cycle. (c) Find the efficiency directly from the results of part (b), and compare with the value calculated from Eq. (20.14).



EXECUTE: (a) With $T_H = (227 + 273.15)\ \text{K} = 500\ \text{K}$ and $T_C = (27 + 273.15)\ \text{K} = 300\ \text{K}$, $pV = nRT$ yields

$$V_a = \frac{nRT_H}{p_a} = \frac{(0.200\ \text{mol})(8.314\ \text{J/mol}\cdot\text{K})(500\ \text{K})}{10.0 \times 10^5\ \text{Pa}}$$

$$= 8.31 \times 10^{-4}\ \text{m}^3$$

The volume doubles during the isothermal expansion $a \rightarrow b$:

$$V_b = 2V_a = 2(8.31 \times 10^{-4}\ \text{m}^3) = 16.6 \times 10^{-4}\ \text{m}^3$$

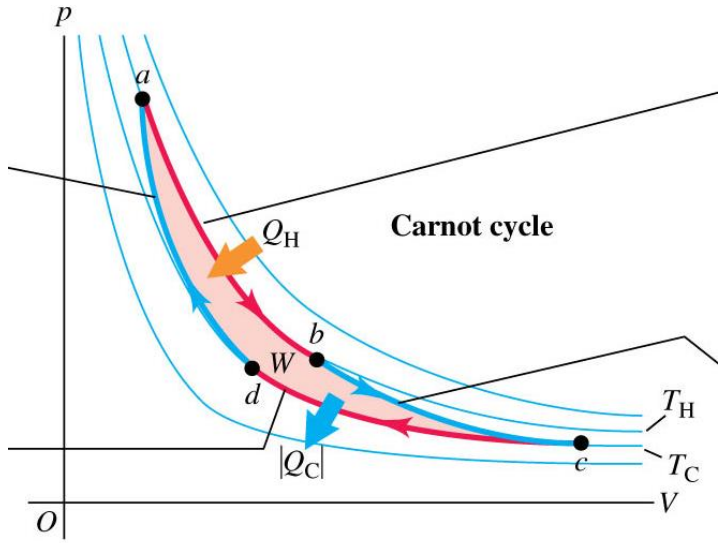
Because the expansion $a \rightarrow b$ is isothermal, $p_aV_a = p_bV_b$, so

$$p_b = \frac{p_aV_a}{V_b} = 5.00 \times 10^5\ \text{Pa}$$

The Carnot Cycle

EXAMPLE 20.3 ANALYZING A CARNOT ENGINE II

Suppose 0.200 mol of an ideal diatomic gas ($\gamma = 1.40$) undergoes a Carnot cycle between 227°C and 27°C, starting at $p_a = 10.0 \times 10^5$ Pa at point a in the pV-diagram of Fig. 20.13. The volume doubles during the isothermal expansion step $a \rightarrow b$. (a) Find the pressure and volume at points a, b, c, and d. (b) Find Q, W, and ΔU for each step and for the entire cycle. (c) Find the efficiency directly from the results of part (b), and compare with the value calculated from Eq. (20.14).



For the adiabatic expansion $b \rightarrow c$, we use the equation $T_H V_b^{\gamma-1} = T_C V_c^{\gamma-1}$ that follows Eq. (20.12) as well as the ideal-gas equation:

$$V_c = V_b\left(\frac{T_H}{T_C}\right)^{1/(\gamma-1)} = (16.6 \times 10^{-4}\ \text{m}^3)\left(\frac{500\ \text{K}}{300\ \text{K}}\right)^{2.5}$$

$$= 59.6 \times 10^{-4}\ \text{m}^3$$

$$p_c = \frac{nRT_C}{V_c} = \frac{(0.200\ \text{mol})(8.314\ \text{J/mol}\cdot\text{K})(300\ \text{K})}{59.6 \times 10^{-4}\ \text{m}^3}$$

$$= 0.837 \times 10^5\ \text{Pa}$$

For the adiabatic compression $d \rightarrow a$, we have $T_C V_d^{\gamma-1} = T_H V_a^{\gamma-1}$ and so

$$V_d = V_a\left(\frac{T_H}{T_C}\right)^{1/(\gamma-1)} = (8.31 \times 10^{-4}\ \text{m}^3)\left(\frac{500\ \text{K}}{300\ \text{K}}\right)^{2.5}$$

$$= 29.8 \times 10^{-4}\ \text{m}^3$$

$$p_d = \frac{nRT_C}{V_d} = \frac{(0.200\ \text{mol})(8.314\ \text{J/mol}\cdot\text{K})(300\ \text{K})}{29.8 \times 10^{-4}\ \text{m}^3}$$

$$= 1.67 \times 10^5\ \text{Pa}$$

The Carnot Cycle

EXAMPLE 20.3 ANALYZING A CARNOT ENGINE II

Suppose 0.200 mol of an ideal diatomic gas ($\gamma = 1.40$) undergoes a Carnot cycle between 227°C and 27°C, starting at $p_a = 10.0 \times 10^5$ Pa at point a in the pV-diagram of Fig. 20.13. The volume doubles during the isothermal expansion step $a \rightarrow b$. (a) Find the pressure and volume at points a, b, c, and d. (b) Find Q, W, and ΔU for each step and for the entire cycle. (c) Find the efficiency directly from the results of part (b), and compare with the value calculated from Eq. (20.14).

(b) For the isothermal expansion $a \rightarrow b$, $\Delta U_{ab} = 0$. From Eq. (20.10),

$$W_{ab} = Q_H = nRT_H \ln\frac{V_b}{V_a}$$

$$= (0.200\ \text{mol})(8.314\ \text{J/mol}\cdot\text{K})(500\ \text{K})(\ln 2) = 576\ \text{J}$$

For the adiabatic expansion $b \rightarrow c$, $Q_{bc} = 0$. From the first law of thermodynamics, $\Delta U_{bc} = Q_{bc} - W_{bc} = -W_{bc}$; the work W_{bc} done by the gas in this adiabatic expansion equals the negative of the change in internal energy of the gas. From Eq. (19.13) we have $\Delta U = nC_V\Delta T$, where $\Delta T = T_C - T_H$. Using $C_V = 20.8\ \text{J/mol}\cdot\text{K}$ for an ideal diatomic gas, we find

$$W_{bc} = -\Delta U_{bc} = -nC_V(T_C - T_H) = nC_V(T_H - T_C)$$

$$= (0.200\ \text{mol})(20.8\ \text{J/mol}\cdot\text{K})(500\ \text{K} - 300\ \text{K}) = 832\ \text{J}$$

For the isothermal compression $c \rightarrow d$, $\Delta U_{cd} = 0$; Eq. (20.11) gives

$$W_{cd} = Q_C = nRT_C \ln\frac{V_d}{V_c}$$

$$= (0.200\ \text{mol})(8.314\ \text{J/mol}\cdot\text{K})(300\ \text{K})\left(\ln\frac{29.8 \times 10^{-4}\ \text{m}^3}{59.6 \times 10^{-4}\ \text{m}^3}\right)$$

$$= -346\ \text{J}$$

For the adiabatic compression $d \rightarrow a$, $Q_{da} = 0$ and

$$W_{da} = -\Delta U_{da} = -nC_V(T_H - T_C) = nC_V(T_C - T_H)$$

$$= (0.200\ \text{mol})(20.8\ \text{J/mol}\cdot\text{K})(300\ \text{K} - 500\ \text{K}) = -832\ \text{J}$$

The Carnot Cycle

EXAMPLE 20.3 ANALYZING A CARNOT ENGINE II

Suppose 0.200 mol of an ideal diatomic gas ($\gamma = 1.40$) undergoes a Carnot cycle between 227°C and 27°C, starting at $p_a = 10.0 \times 10^5$ Pa at point a in the pV-diagram of Fig. 20.13. The volume doubles during the isothermal expansion step $a \rightarrow b$. (a) Find the pressure and volume at points a, b, c, and d. (b) Find Q, W, and ΔU for each step and for the entire cycle. (c) Find the efficiency directly from the results of part (b), and compare with the value calculated from Eq. (20.14).

Process	Q	W	ΔU
$a \rightarrow b$	576 J	576 J	0
$b \rightarrow c$	0	832 J	−832 J
$c \rightarrow d$	−346 J	−346 J	0
$d \rightarrow a$	0	−832 J	832 J
Total	230 J	230 J	0

(c) From the above table, $Q_H = 576$ J and the total work is 230 J. Thus

$$e = \frac{W}{Q_H} = \frac{230\ \text{J}}{576\ \text{J}} = 0.40 = 40\%$$

We can compare this to the result from Eq. (20.14),

$$e = \frac{T_H - T_C}{T_H} = \frac{500\ \text{K} - 300\ \text{K}}{500\ \text{K}} = 0.40 = 40\%$$

The Carnot Refrigerator

- The important feature of the Carnot cycle is that it is reversible, because at every stage, it is almost in equilibrium.
- Hence, the entire cycle may be reversed, converting the engine into a refrigerator.
- Coefficient of performance of the Carnot refrigerator

$$K = \frac{|Q_C|}{|Q_H| - |Q_C|} = \frac{|Q_C|/|Q_H|}{1 - |Q_C|/|Q_H|}$$

$$|Q_C|/|Q_H| = T_C/T_H$$

Coefficient of performance of a Carnot refrigerator

$$K_{\text{Carnot}} = \frac{T_C}{T_H - T_C}$$

T_C: Temperature of cold reservoir

T_H: Temperature of hot reservoir

(20.15)

- When the temperature difference $T_H - T_C$ is small, K is much larger than unity; in this case a lot of heat can be "pumped" from the lower to the higher temperature with only a little expenditure of work.

The Carnot Refrigerator

EXAMPLE 20.4 ANALYZING A CARNOT REFRIGERATOR

If the cycle described in Example 20.3 is run backward as a refrigerator, what is its coefficient of performance?

SOLUTION

IDENTIFY and SET UP: This problem uses the ideas of Section 20.3 (for refrigerators in general) and the above discussion of Carnot refrigerators. Equation (20.9) gives the coefficient of performance K of *any* refrigerator in terms of the heat Q_C extracted from the cold reservoir per cycle and the work W that must be done per cycle.

EXECUTE: In Example 20.3 we found that in one cycle the Carnot engine rejects heat $Q_C = -346$ J to the cold reservoir and does work $W = 230$ J. When run in reverse as a refrigerator, the system extracts heat $Q_C = +346$ J from the cold reservoir while requiring a work input of $W = -230$ J. From Eq. (20.9),

$$K = \frac{|Q_C|}{|W|} = \frac{346\ \text{J}}{230\ \text{J}} = 1.50$$

Because this is a Carnot cycle, we can also use Eq. (20.15):

$$K = \frac{T_C}{T_H - T_C} = \frac{300\ \text{K}}{500\ \text{K} - 300\ \text{K}} = 1.50$$

EVALUATE: Equations (20.14) and (20.15) show that e and K for a Carnot cycle depend only on T_H and T_C, and we don't need to calculate Q and W. For cycles containing irreversible processes, however, these two equations are not valid, and more detailed calculations are necessary.

The Kelvin Temperature Scale

- The thermal efficiency of a Carnot engine operating between two heat reservoirs at temperatures and is independent of the nature of the working substance and depends only on the temperatures.
- Therefore the ratio Q_C/Q_H is the same for all Carnot engines operating between two given temperatures T_H and T_C .

$$e = \frac{Q_H + Q_C}{Q_H} = 1 + \frac{Q_C}{Q_H} \quad \Rightarrow \quad e_{\text{Carnot}} = 1 - \frac{T_C}{T_H} = \frac{T_H - T_C}{T_H}$$

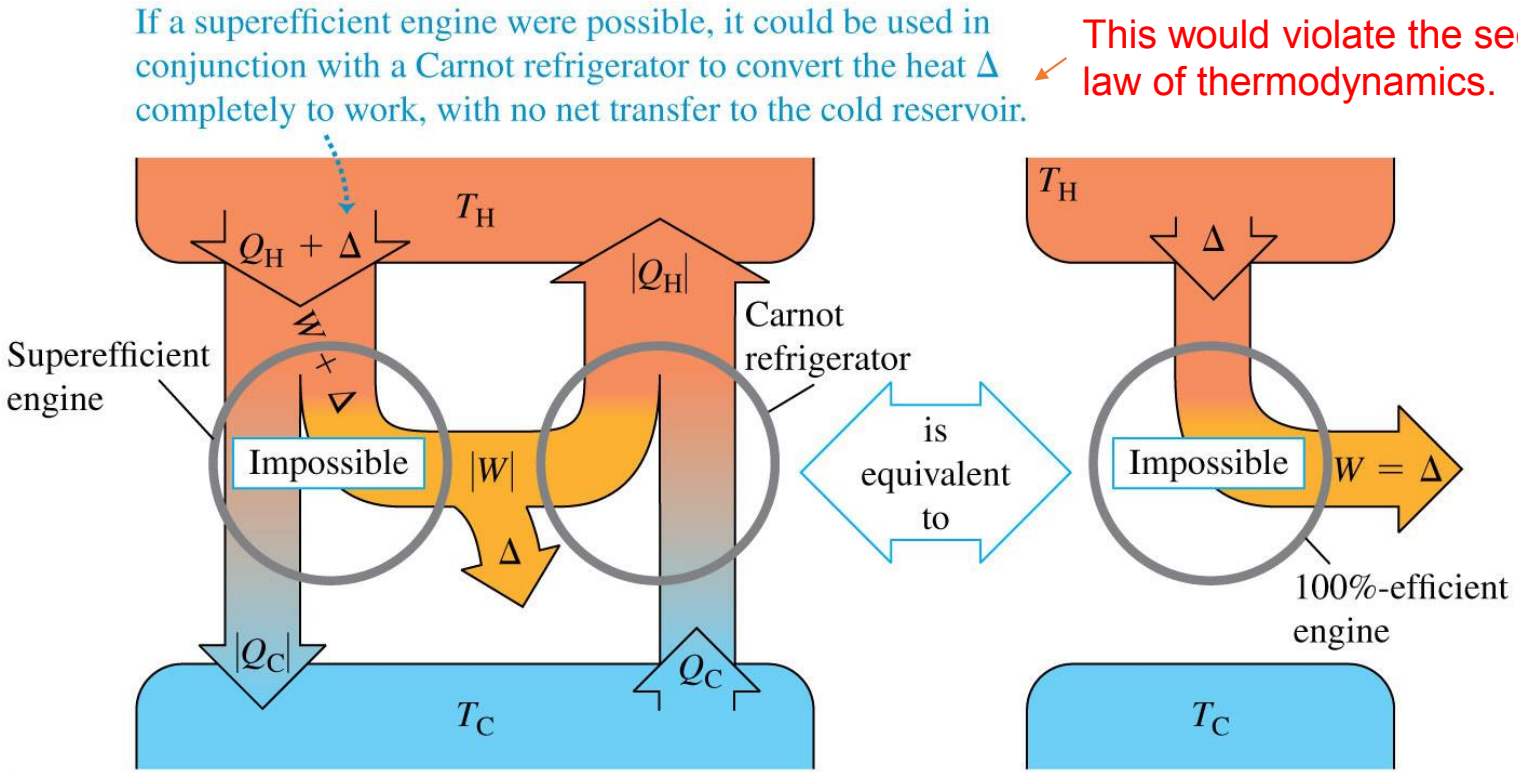
- Kelvin proposed that we define the ratio of the temperatures, to be equal to the magnitude of the ratio of the quantities of heat absorbed and rejected:

$$\frac{T_C}{T_H} = \frac{|Q_C|}{|Q_H|} = -\frac{Q_C}{Q_H} \qquad \text{(definition of Kelvin temperature)} \qquad (20.16)$$

- Thus the Kelvin temperature scale is truly absolute.

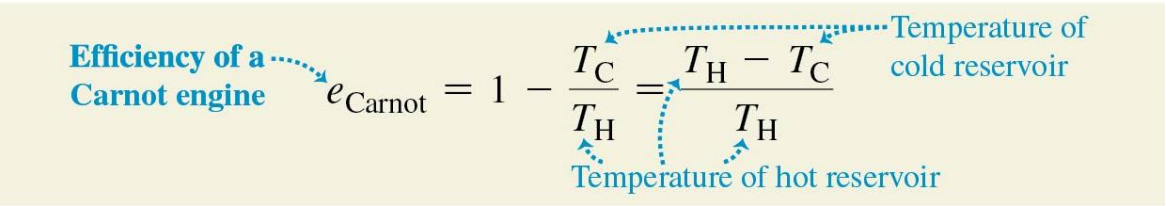
The Carnot Cycle and the Second Law

- No engine can be more efficient than a Carnot engine operating between the same two temperatures.



The Carnot Cycle and the Second Law

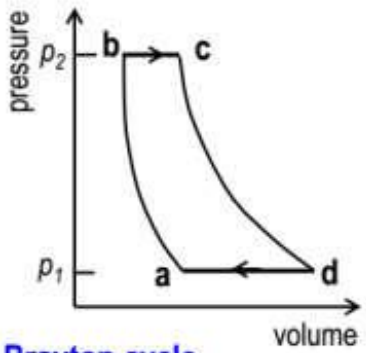
- All Carnot engines operating between the same two temperatures have the same efficiency, irrespective of the nature of the working substance.
- Design target/guidance



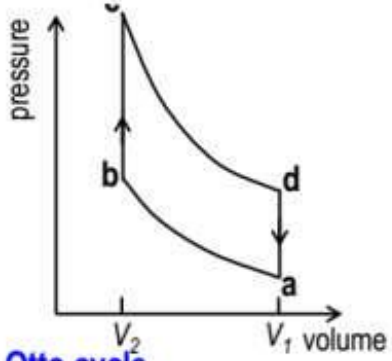
(20.14)

- However, the Carnot cycle is unfortunately impractical due to requirement of reversible isothermal process, which is too slow for practical applications.
- Reversible adiabatic processes on the other hand are easy to implement.

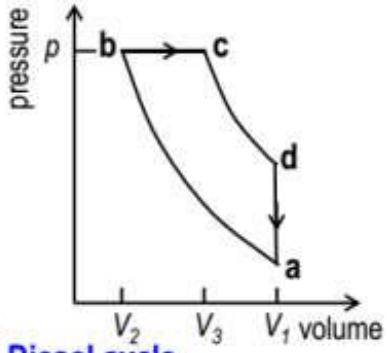
Other Heat Engine Cycles



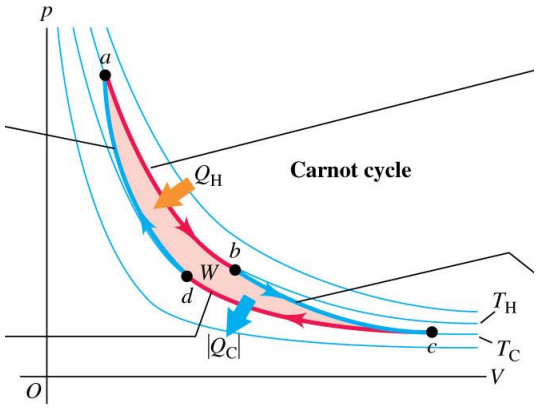
Brayton cycle
(gas turbine engine)

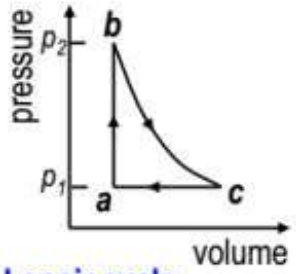


Otto cycle
(petrol engine)

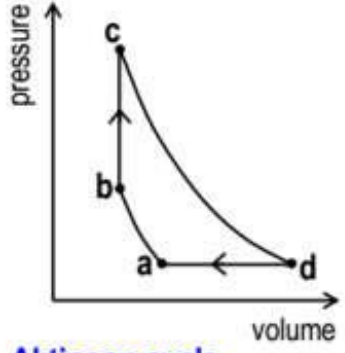


Diesel cycle
(Diesel engine)

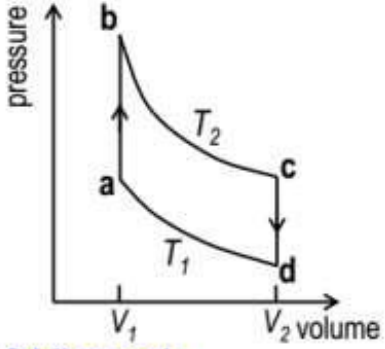




Lenoir cycle
(pulsed jet engine)



Aktinson cycle
(Toyota Prius hybrid car)



Stirling cycle
(Stirling engine)

Summary

- The First Law of Thermodynamics has been studied.
  - Basic concepts and principles are discussed with detailed analysis and examples
    1. **Directions of Thermodynamic Processes**
    2. **Heat Engines**
    3. **Internal-Combustion Engines**
    4. **Refrigerators**
    5. **The Second Law of Thermodynamics**
    6. **Carnot Cycles**
- Textbook
  - Chapter 20: The Second Law of Thermodynamics